Edge AI Smart Battery Management System with Intelligent UPS

Group 1

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Introduction

- **Widespread Applications**: Lithium-ion battery packs are extensively used in electric vehicles (EVs), renewable energy storage, robotics, and industrial backup systems.
- **Conventional Battery Management System (BMS) Limitations**: Provides only basic hardware protection (over/under-voltage, over-current) with very limited insight into true battery health.
- **Intelligent BMS Upgrade**: Integrates Edge AI, predictive analytics, and smart UPS capability directly on an advanced ESP32-S3 platform.
- **Real-Time Intelligence**: Enables accurate State-of-Charge(SOC), State-of-Health(SOH), and Remaining Useful Life(RUL) estimation alongside proactive on-device fault detection.

Problem Statement

- **Basic Protection Only**: Conventional BMS acts mainly as a simple hardware cutoff device without diagnostic intelligence or forecasting abilities.
- **Lack of Predictive Insights**: Unable to accurately evaluate cell degradation or predict Remaining Useful Life (RUL) under varying operating conditions.
- **Financial & Safety Risks**: Unprecedented cell aging leads to premature or delayed battery replacement, increasing overall lifecycle costs.

Proposed Need

An Edge AI-based Intelligent BMS for sophisticated predictive analytics, real-time telemetry, and autonomous UPS switching.

Objectives

- **Hardware & Edge Platform**: Build a standalone, operational Edge AI Smart Battery Management System based on the ESP32-S3 microcontroller.
- **Real-Time Parameter Sensing**: Implement high-precision monitoring for cell voltage, load current, and pack temperature parameters.
- **Predictive AI Algorithms**: Deploy lightweight machine learning models for accurate SOC, SOH, and RUL estimation.
- **Autonomous Protection**: Enable automated on-device fault detection coupled with immediate, glitch-free UPS backup mode switching.
- **Visualization Dashboard**: Develop an intuitive, cross-platform Flutter monitoring dashboard featuring analytical graphs and real-time alerts.

Literature Review

No.	Authors	Year	Paper title	Key technical inferences
1	Rui Xiong et al.	2018	Critical Review on the Battery State of Charge Estimation Methods for Electric Vehicles	Reviews lookup-table, coulomb-counting, model-based, and data-driven SOC methods. Reports errors below 5% for an ECM approach, below 2% for multi-scale EKF, and below $\pm 1\%$ for an impedance-model approach. Highlights training-data coverage, model accuracy, cell inconsistency, and aging as challenges; motivates data-driven SOC estimation and lightweight deployment.
2	Chemali et al.	2018	Long Short-Term Memory Networks for Accurate State-of-Charge Estimation of Li-ion Batteries	Maps voltage, current, and temperature directly to SOC with an LSTM-RNN, without an explicit battery model or Kalman filter. A 500-node LSTM reported 0.573% MAE at 1000 time steps under fixed temperature and 1.606% in a 10–25°C ramp test. Results used one NCA cell; operating conditions and initialization affect error, and embedded resource needs require separate validation.

Literature Review (Contd.)

No.	Authors	Year	Paper title	Key technical inferences
3	G. Crocioni et al.	2020	Li-Ion Batteries Parameter Estimation With Tiny Neural Networks Embedded on Intelligent IoT Microcontrollers	Compares CNN, LSTM, GRU, and hybrid models for maximum releasable capacity using 1,462 NASA observations. CNN-GRU reports maximum RMSE 0.0488 and MAE 0.0414. Embedded deployment is evaluated with STM32Cube.AI and CNN inference benchmarked against TFLM; 8-bit quantization cuts weights and RAM by over 50% with an accuracy trade-off. Larger errors on a held-out battery motivate battery-wise testing and ESP32-S3 validation.
4	P. Venugopal et al.	2021	Analysis of Optimal Machine Learning Approach for Battery Life Estimation of Li-Ion Cell	Uses the NASA Li-ion battery ageing dataset to compare ML models for SOH and RUL estimation. DNN reports SOH estimation error within $\pm 5\%$, while LSTM estimates RUL within ± 10 cycles. Demonstrates hardware-based implementation using Raspberry Pi 3, motivating further work on lightweight TinyML models for real-time battery-health estimation on resource-constrained microcontrollers such as ESP32-S3.

Literature Review (Contd.)

No.	Authors	Year	Paper title	Key technical inferences
5	H. M. Hussein et al.	2024	Comparative Study-Based Data-Driven Models for Lithium-Ion Battery State-of-Charge Estimation	Compares LR, NN, AE, Transformer, LSTM, and Random Forest Regression using measured $LiFePO_4$ pack data. RFR reports MSE 0.0002765, MAE 0.0007379, and $R^2 = 0.9999996$. The study shows model complexity alone does not determine performance; model size, computation, and generalization matter for embedded deployment.
6	S. Giazitzis et al.	2024	A Case Study of a Tiny Machine Learning Application for Battery State-of-Charge Estimation	Deploys ANN and CNN models on a microcontroller using post-training quantization. 16-bit quantization reduces weight memory by over 50% with minimal accuracy loss; ANN uses 20,246 inference cycles and 12.66 kB, with lower compute and memory cost than CNN. Aging increases SOC errors, so deployment must balance accuracy, quantization loss, RAM, and inference cost.

Literature Review (Contd.)

No.	Authors	Year	Paper title	Key technical inferences
7	P. Li et al.	2025	Applying Neural Network to Health Estimation and Lifetime Prediction of Lithium-Ion Batteries	Surveys ANN architectures for SOH estimation and RUL prediction from aging data. Shallow ANNs use fewer resources but may not capture complex degradation; deep or hybrid ANNs capture richer patterns at higher computational and memory cost. Recommends evaluating prediction error together with RAM, model size, and inference latency for embedded use.
8	G. Giazitzis et al.	2026	Embedded Strategy for Battery Module States Estimation Using Tiny Machine Learning Models	Uses neural-network projection with fine-tuning to compress models for TinyML. Projected LSTM reports SOC MAE 0.0114 with a 4.0 kB model; projected CNN-GRU reports SOH MAE 0.0060 with a 3.4 kB model using ICA features. Compression may increase RMSE, and GPU latency alone does not establish microcontroller performance.

Proposed Methodology

- **Data Acquisition**: Capture real-time cell voltage, current, and temperature using high-precision embedded sensors.
- **Battery Monitoring**: Implement parameter estimation and multi-layer hardware safety protection algorithms.
- **Edge AI Modeling**: Deploy lightweight machine learning models trained on historical cycling data for SOH, RUL, and thermal fault prediction.
- **On-Device Inference**: Perform ML inference directly on an ESP32-S3 processor for deterministic, low-latency execution.
- **Smart Management**: Facilitate intelligent UPS backup power switching, wireless telemetry logging, and Flutter dashboard monitoring

Block Diagram

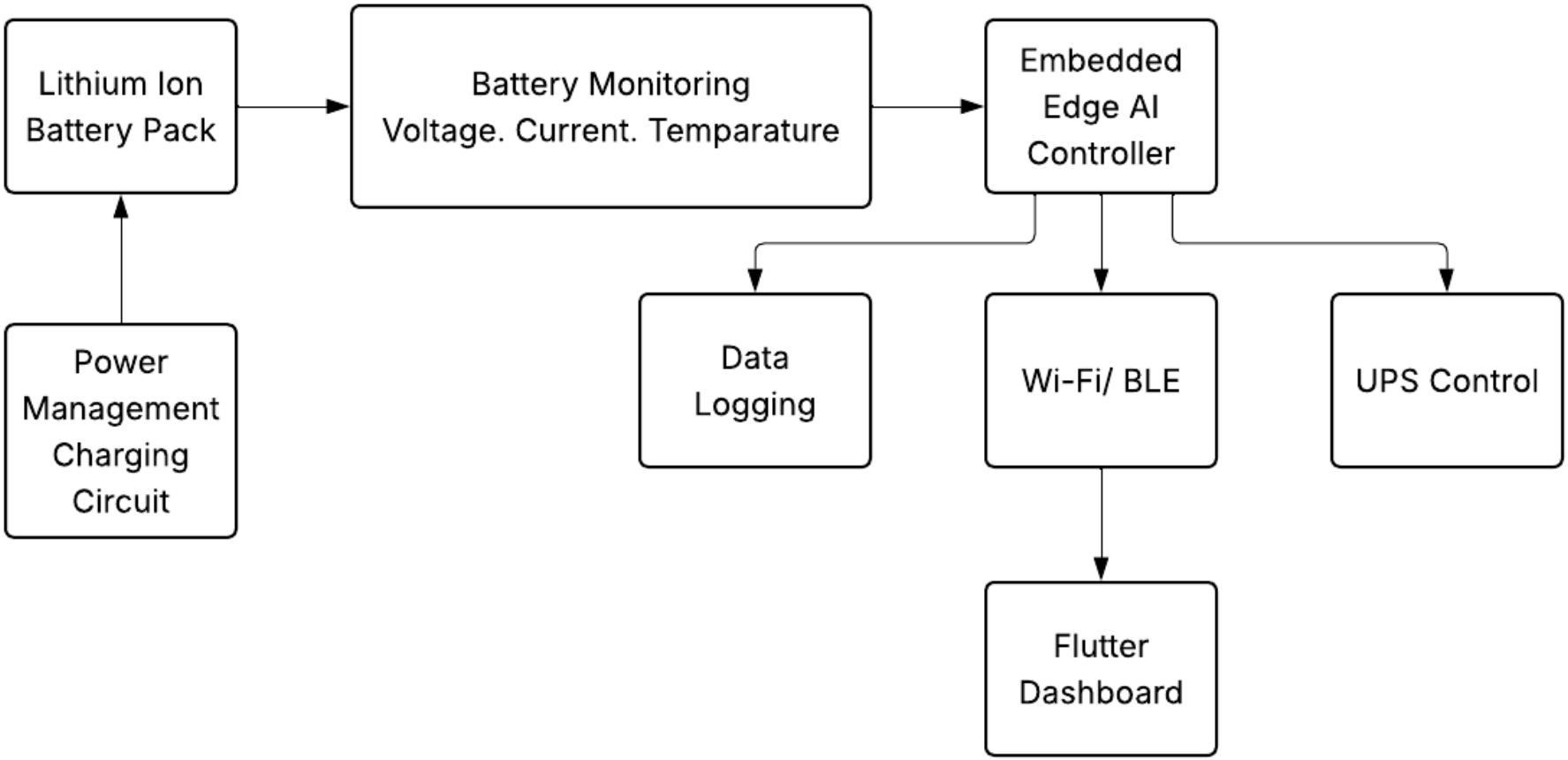


Figure 1: Block Diagram of Edge AI Smart Battery Management System with Intelligent UPS

Prepared Battery Dataset Sample

A	B	C	D	E	F	G	H
battery_id	cycle	t_s	voltage	current	temperature	soc	cycle_capacity_Ah
B0005	1	0	4.191491808	-0.00490158920	24.33003389	100	1.856487421
B0005	1	16.781	4.190749068	-0.00147800555	24.32599342	99.99919909	1.856487421
B0005	1	35.703	3.974870912	-2.012528324	24.38908513	99.71409505	1.856487421
B0005	1	53.781	3.951716708	-2.013979362	24.54475232	99.16952418	1.856487421
B0005	1	71.922	3.934352489	-2.011143596	24.7313852	98.62324347	1.856487421
B0005	1	90.094	3.920058442	-2.013006628	24.9098164	98.07616149	1.856487421
B0005	1	108.281	3.90790351	-2.014399657	25.10588415	97.52818491	1.856487421
B0005	1	126.453	3.897035722	-2.011602885	25.31701856	96.98085111	1.856487421
B0005	1	144.641	3.887476581	-2.018014688	25.5094231	96.4325435	1.856487421
B0005	1	162.844	3.878958592	-2.013134754	25.70360345	95.88357509	1.856487421
B0005	1	181.016	3.871015985	-2.013162202	25.88720819	95.33620126	1.856487421
B0005	1	199.219	3.863479677	-2.013030234	26.07781486	94.7879079	1.856487421
B0005	1	217.391	3.856660507	-2.013710499	26.25431462	94.24047375	1.856487421
B0005	1	235.594	3.850013579	-2.014150137	26.43671069	93.6919532	1.856487421
B0005	1	253.75	3.843841905	-2.013463983	26.61127891	93.14488242	1.856487421
B0005	1	271.984	3.837687976	-2.012645836	26.78134765	92.59566657	1.856487421
B0005	1	290.141	3.831938253	-2.013409301	26.95745885	92.04877743	1.856487421
B0005	1	308.359	3.826414443	-2.0131455	27.12980014	91.49998286	1.856487421
B0005	1	326.5	3.821095734	-2.012041289	27.30582408	90.95369348	1.856487421
B0005	1	344.75	3.815896665	-2.014463253	27.46481668	90.40394181	1.856487421
B0005	1	362.906	3.810696822	-2.01314003	27.61796452	89.8568725	1.856487421

Figure 2: Representative rows from the prepared battery dataset used for SOC and SOH model training.

Design Formulation and Progress

Hardware

- 3S Li-ion pack: 9–12.6 V; cell taps sensed on GPIO 4, 5, and 6.
- INA226 measures current, bus voltage, and power over I^2C (GPIO 8/9).
- Three 10 kΩ NTCs monitor cell temperatures on GPIO 1, 2, and 7.
- ESP32-S3 provides local processing, Wi-Fi, and USB debugging.

Software

- 1 Hz loop samples sensors, calibrates readings, estimates SOC, and reports data.
- Serial commands support logging, cycle tracking, and CSV output.
- Monitoring includes cell imbalance and load activity detection.
- Flutter app presents live status, analytics, predictions, alerts, and settings.

Design Formulation and Progress (Contd.)

- **Hardware and firmware — Complete.**
  ESP32-S3 prototype integrates three cell taps, INA226 current sensing, and three NTCs. Firmware samples and calibrates readings for local monitoring.
- **ADC calibration — In progress.**
  Quadratic fits improve voltage accuracy; Tap 3 still needs recalibration.
- **NASA dataset Preprocessing — Complete.**
  Cycling records are cleaned and organized as time series with voltage, current, temperature, SOC, and capacity fields.
- **SOC models — Complete.**
  RF, MLP, and LSTM were compared with Coulomb counting using battery-level splits; Fine-Tuned MLP preferred for edge-deployment.
- **SOH/RUL models — In progress.**
  Aging- and capacity-based estimators are under development; evaluation and optimization remain before embedded deployment.
- **Flutter app — Complete.**
  The 11-screen app displays battery status, telemetry, analytics, predictions, alerts, and settings.

Results and Discussion

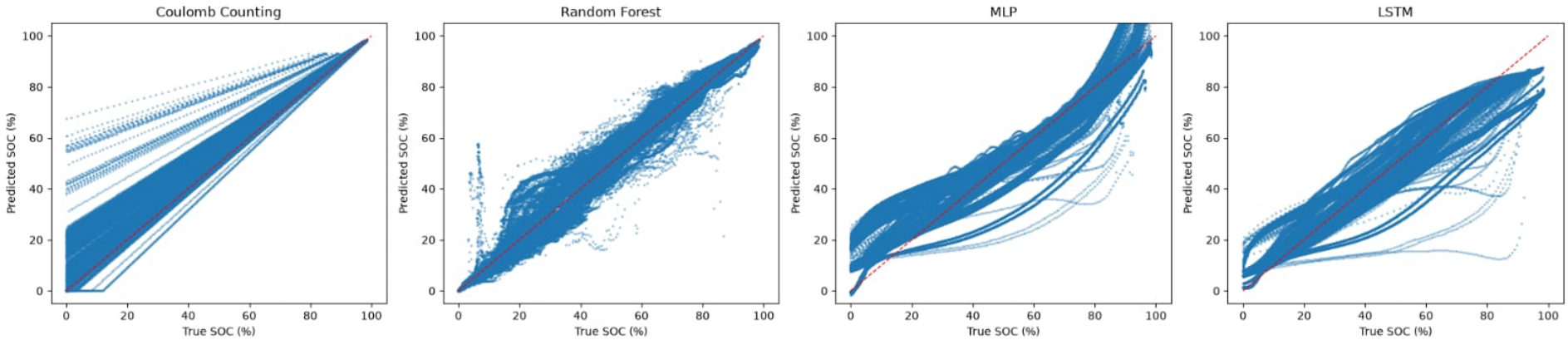
Model	MAE (%)	RMSE (%)	R^2
Coulomb counting	4.266	7.540	0.9273
Random Forest	3.587	5.152	0.9661
XGBoost	3.964	5.401	0.9627
Simplified MLP / DNN	7.591	9.693	0.8799
Fine-tuned MLP / DNN	**4.647**	**6.274**	**0.9497**
LSTM	7.070	9.275	0.8900

Table 5. SOC Model Performance Comparison

Key Finding

The fine-tuned MLP is preferred for edge deployment because its lightweight design, low complexity, and small footprint suit microcontrollers. It achieves MAE of 4.647%, RMSE of 6.274%, and R^2 of 0.9497.

SOC Model Comparison - MAE per Battery



(a) Coulomb counting (b) Random Forest (c) MLP (d) LSTM

Figure 3: Actual versus predicted State of Charge (SOC) for four estimation methods.

Fine-Tuned MLP: Actual vs. Predicted SOC

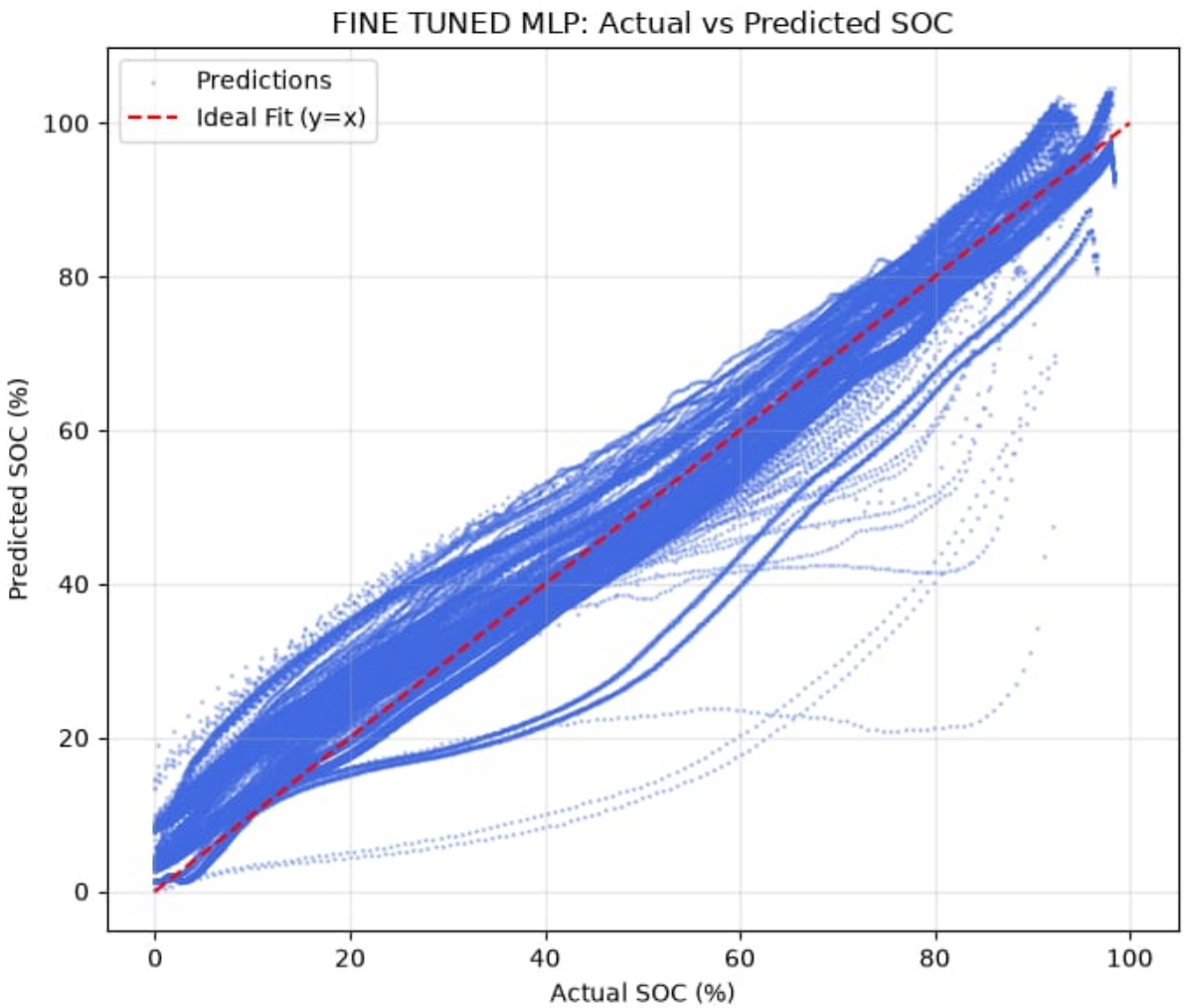
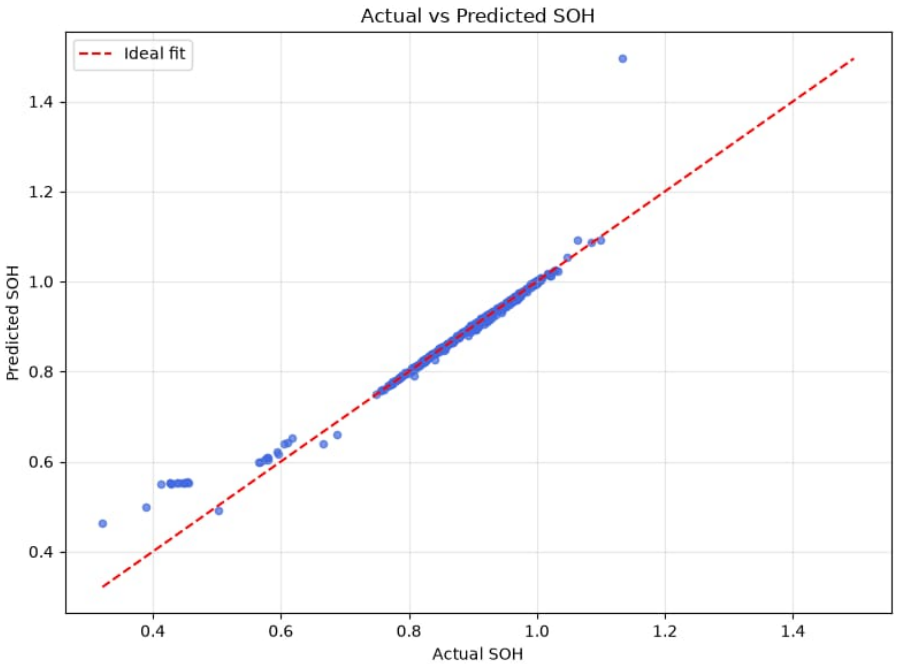
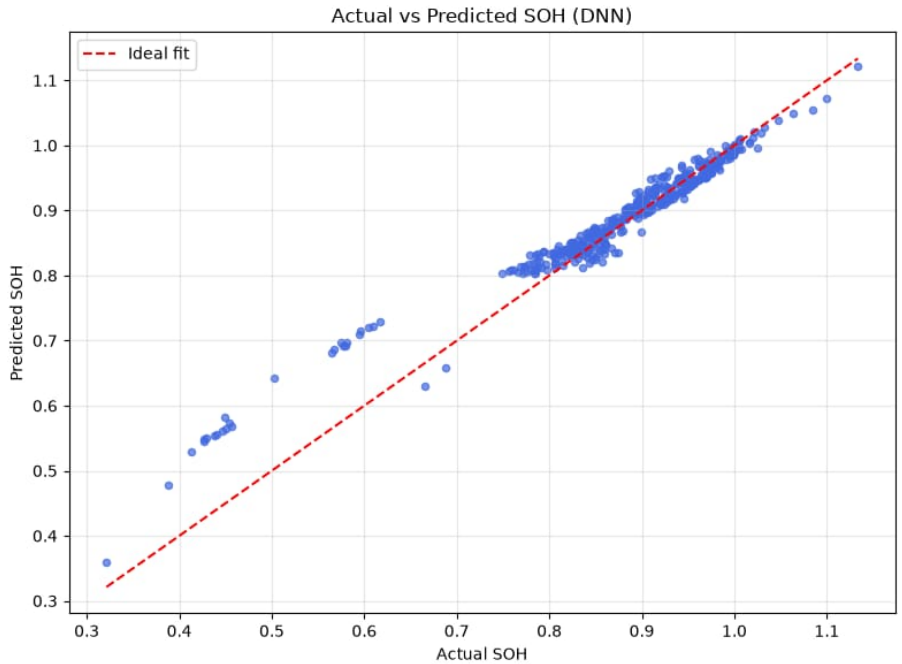


Figure 4: Actual versus predicted SOC for the fine-tuned MLP model.

SOH Estimation: Actual vs. Predicted



(a) XGBoost based SOH estimation



(b) DNN-based SOH estimation

Figure 5: Actual versus Predicted SOH; dashed lines indicate ideal fit.

SOH Estimation: Model Comparison

Model	MAE (%)	RMSE (%)	R^2
XGBoost	0.0072	0.0270	0.9494
DNN	**0.0194**	**0.0320**	**0.9287**

Table 6. SOH Model Performance Comparison

Key Finding

DNN is preferred for edge deployment because compared to XGBoost it is lightweight , less complex, and have small footprint suit microcontrollers. It achieves MAE of 0.0194%, RMSE of 0.0320%, and R^2 of 0.9287.

Discussion: Limitations and Deployment Challenges

- **Dataset Limitations**: NASA data contains four discrete current levels (0, -1, -2, and -4 A), limiting real-world variable-load representation.
- **Voltage–SOC Dependence**: Hysteresis (0.3–0.5 V), temperature, and aging affect the voltage–SOC relationship.
- **Model Generalization**: Test batteries B0038 and B0054 showed systematic prediction errors, indicating possible limitations across battery units.
- **Limitations of Conventional SOC Estimation**: Coulomb counting suffers from cumulative integration errors and initial SOC uncertainty, motivating ML-based estimation. However, ML accuracy depends on reference-label quality.
- **Edge Deployment and Validation**: The fine-tuned MLP (SOC) and DNN (SOH) were selected for deployment feasibility. ESP32-S3 inference and validation under variable loads and temperatures remain pending.

Hardware Setup

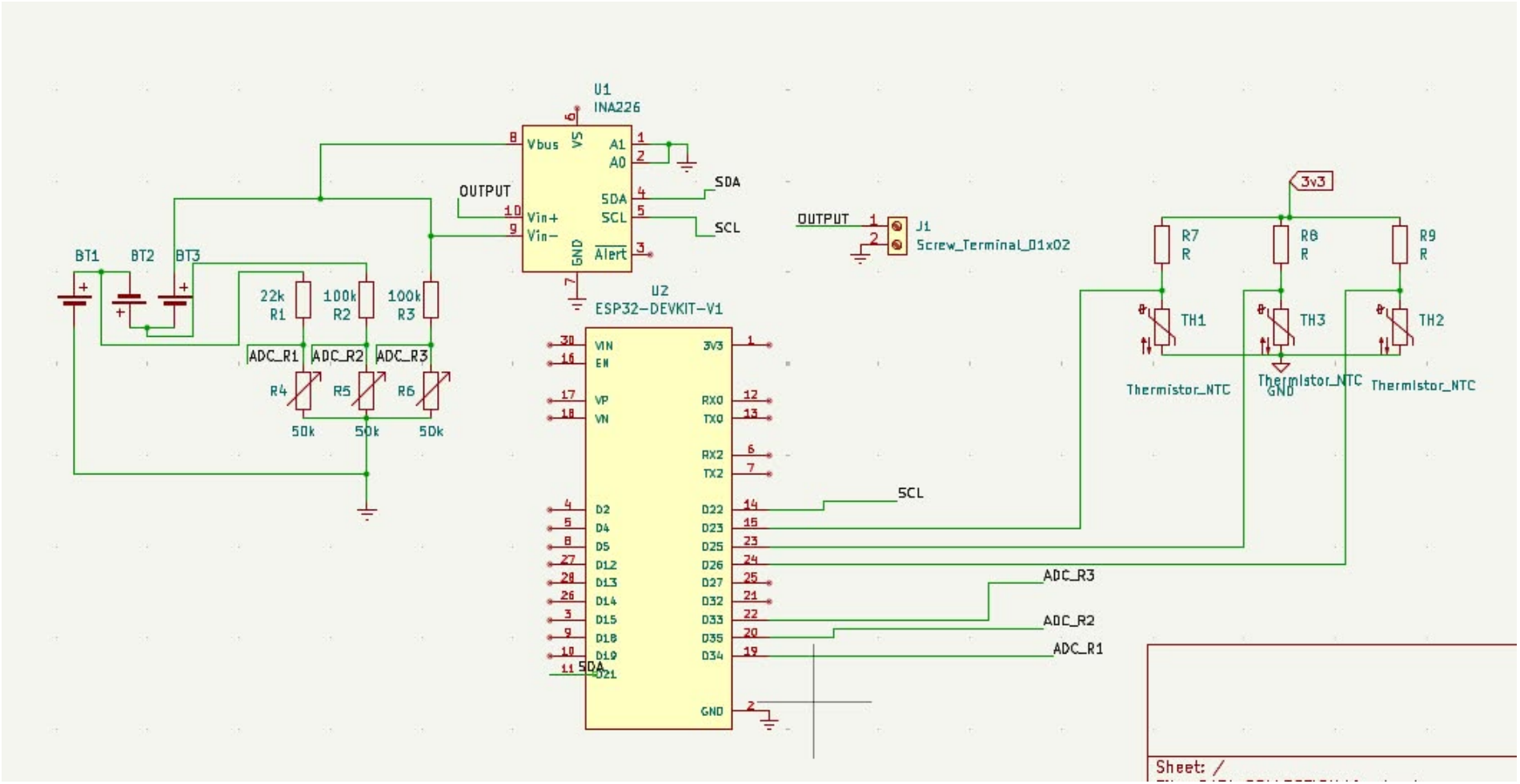


Figure 6: Hardware setup of the Edge AI Smart Battery Management System.

System Implementation and Functional Features

ESP32-S3 Firmware Capabilities

- Three-cell voltage monitoring with quadratic calibration.
- Load-current measurement using INA226.
- Temperature sensing with three NTC thermistors.
- NASA-compatible CSV output and serial command interface.
- Cycle tracking with capacity integration.
- Cell-imbalance detection.
- SoC estimation using a linear method.

Flutter App Features

- Real-time dashboard with circular SoC indicator.
- Historical voltage, current, temperature, SoC, SOH, and power analytics.
- AI prediction screen with degradation forecast.
- Battery details with cell-level monitoring.
- Severity-level alerts and device-management settings.
- Dark and light themes.
- OTA update interface (UI ready).

Prototype Performance Metrics

Metric	Target	Achieved
SOC MAE	$< 5\%$	3.59%
ADC Accuracy	< 50 mV	31.5 mV (Tap 1); 56 mV (Tap 2)
Voltage Resolution	12-bit	12-bit
Current Sensing	± 0.8 A	± 0.8 A
Temperature Sensing	3 cells	3 cells
Mobile Screens	8+	11

Table 7. Hardware and Dashboard Performance Metrics

Conclusion

The prototype combines embedded sensing, Edge AI, and mobile visualization. UPS/OTA integration and end-to-end validation remain in progress toward reliable deployment.

Individual Contributions & Teamwork

Member	Responsibilities
Harikesh OP	Embedded firmware, ML models, and hardware integration.
Amaya Promod PV	Literature review, NASA dataset study, and data augmentation.
Nandana Jayachandran	Dataset analysis and data augmentation.
Pooja K	Documentation and supporting tasks.

Table 6. Team Member Contributions

Teamwork & Collaboration

- Regular progress meetings and code reviews.
- Collaborative debugging and knowledge sharing.
- Coordinated hardware–software integration.

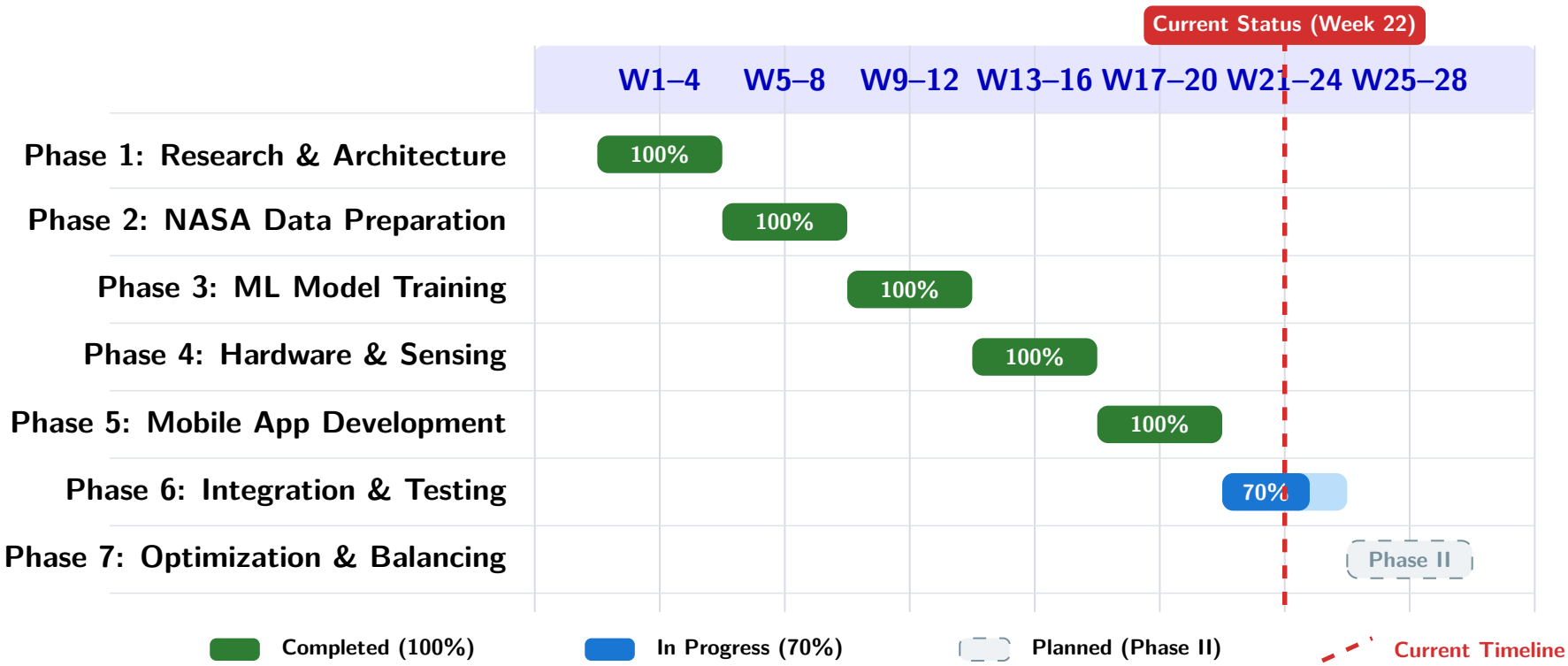
Tools & Technologies

Embedded: C++, Arduino, PlatformIO, ESP32/S3
ML/Data: Python, pandas, scikit-learn, TensorFlow/Keras, NumPy
Mobile: Flutter, Dart, Provider, fl_chart
Hardware: INA226, NTC thermistors, voltage dividers
Tools: Git, Jupyter, VS Code, Arduino IDE

Work Plan

- **Phase 1 – Research & Design (Weeks 1–4, complete):** Literature review, NASA dataset study, system architecture, and hardware selection.
- **Phase 2 – Data Preparation (Weeks 5–8, complete):** Dataset extraction, cleaning, preprocessing, analysis, and train/test split.
- **Phase 3 – ML Development (Weeks 9–12, complete):** Coulomb counting baseline; RF, MLP, and LSTM training, evaluation, and comparison.
- **Phase 4 – Hardware (Weeks 13–16, complete):** ESP32/S3 firmware, ADC calibration, INA226, NTC sensing, and NASA data mode.
- **Phase 5 – Mobile App (Weeks 17–20, complete):** Flutter architecture, dashboard, analytics, AI predictions, settings, and themes.
- **Phase 6 – Integration & Testing (Weeks 21–24, in progress):** Wi-Fi, SD logging, OTA, system tests, fixes, and optimization.

Project Schedule & Progress (Gantt Chart)



Progress Summary

Phases 1–5 are 100% complete. Phase 6 (System Integration & Validation) is actively underway at 70% completion, targeting end-to-end telemetry, UPS changeover, and field testing prior to Phase II deployment.

Project Budget & Component Cost Breakdown

No.	Component / Module	Specification / Purpose	Qty	Rate (Rs.)	Cost (Rs.)	Status
1	ESP32-S3 DevKit-N16R8	Dual Xtensa LX7, 8MB PSRAM, Wi-Fi	1	650	650	**Procured**
2	INA226 Sensor Module	I^2C Bi-dir Current/Bus Monitor (0.1 Ω)	1	280	280	**Procured**
3	3S Li-ion Battery Pack	18650 Cells (11.1 V nominal, 2600 mAh)	3	250	750	**Procured**
4	NTC Thermistors (10 kΩ)	Cell temperature monitoring ($B = 3950$)	3	30	90	**Procured**
5	Voltage Divider Resistors	Carbon-film resistors & trimpots / caps	1 set	80	80	**Procured**
6	UPS Switching Circuit	Auto-cutover relay, MOSFETs & buck module	1	350	350	**Not Procured**
7	MicroSD Logging Module	SPI interface with 16 GB Class 10 card	1	260	260	**Not Procured**
8	Prototyping & Packaging	Perfboard, terminal blocks, wiring harness	1 set	190	190	**Procured**
Total Budget (Procured: Rs. 2,040 | Pending: Rs. 610):					**Rs. 2,650**	**Within Budget**

Cost Effectiveness

- **Commercial Smart BMS:** Typically Rs. 12,000–25,000 ($150–$300).
- **Proposed Prototype:** Full build within **Rs. 2,650** ($32); **Rs. 2,040** utilized for active hardware to date.

PO11 Alignment (Finance & Management)

- Quadratic ADC calibration compensates for standard carbon resistor tolerances without costly AFEs.
- Phased procurement: UPS and SD logger scheduled for final integration milestone.

Future Activities

- ADC Calibration improvement
- Wi-Fi Integration
- OTA Firmware Updates
- SOH and RUL Models
- Real hardware data collection
- Model Optimization for Edge deployment
- Cascaded SOC–SOH Architecture
- Cell balancing Implementation
- Advanced Protection Features
- Multi-pack Support
- Production Readiness

Project Summary

Objective

Develop an intelligent BMS that protects lithium-ion batteries, predicts battery health using Edge AI, and aims to provide uninterrupted power through an integrated UPS.

Key Achievements

- Hardware Prototype
- Machine Learning Models
- Data Pipeline
- Mobile Application
- Firmware features

Novel Contributions

- End-to-end system
- Comparative ML study
- Data Quality Improvements
- Real Hardware Implementation

Conclusion

The prototype combines embedded sensing, Edge AI, and mobile visualization. UPS/OTA integration and end-to-end validation remain in progress toward reliable deployment.

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Appendix A: NBA Accreditation Criteria

Table A.1: Student Capstone Projects with PO, PSO and Sustainability Mapping

Sl. No.	Project Title	Guide	Type	Domain	SDGs
1	Edge AI Smart Battery Management System with Intelligent UPS	Dr. Sajith K	Product-oriented	Embedded Systems, Edge AI, IoT, and Mobile Application Development	SDG 7, 9, 11

Environment	Safety	Ethics	Cost	Standards	POs & PSOs	POs & PSOs Justification
Low to Moderate impact	Low risk	Low concern	Low	IEEE 802.11 (Wi-Fi), IEC 62660-1, and IEEE 1725	PO1–PO5, PO7, PO9–PO12; PSO1–PSO3	Integrates embedded systems and Edge AI for intelligent battery monitoring, model evaluation, and sustainable battery usage.

Appendix A: NBA Accreditation Criteria (Contd.)

PO Mapping Justification

PO	Justification
PO1	Applied engineering and mathematics to BMS design, quadratic ADC calibration, voltage dividers, and NTC sensing.
PO2	Analyzed conventional BMS limits, NASA data-quality issues, and out-of-distribution battery behavior.
PO3	Designed system architecture and developed ESP32-S3 firmware, ML estimators, and a Flutter app.
PO4	Conducted NASA EDA, hysteresis and aging analysis, and systematic model comparisons.
PO5	Used Python, scikit-learn, TensorFlow, PlatformIO, Flutter, and Git.
PO7	Predictive maintenance supports longer battery life, less waste, and efficient energy use.
PO9	Led firmware development; collaborated on ML and integration through meetings and reviews.
PO10	Prepared project documentation, technical reports, and presentations.
PO11	Managed phased milestones, budget-conscious components, risks, and future resources.
PO12	Learned Flutter, TensorFlow, and ESP32-S3 and iteratively improved the system using research findings.

PSO Mapping Justification

PSO	Justification
PSO1	Hands-on competence in electronic circuits and embedded systems.
PSO2	Proficiency in ML programming and software tools.
PSO3	Application of instrumentation and monitoring techniques for battery management.

Appendix A: NBA Accreditation Criteria (Contd.)

SDG Mapping Justification

SDG	Justification
SDG 7	Promotes efficient battery utilization and supports sustainable energy storage through intelligent battery monitoring.
SDG 9	Encourages technological innovation by integrating embedded systems, Edge AI, and IoT for smart battery management.
SDG 11	Contributes to sustainable and resilient communities by supporting reliable battery storage and improved energy management.

Project Attributes Justification

Attribute	Justification
Type	Develops a product-oriented Smart BMS prototype integrating intelligent battery monitoring and UPS functionality.
Domain	Integrates embedded systems, Edge AI, Wi-Fi-based IoT connectivity, and mobile application development.
Environment	Predictive maintenance can extend battery life, reduce battery waste, and improve energy utilization.
Safety	Battery sensing and safety checks are considered; end-to-end protection validation is ongoing.
Ethics	Uses the public NASA dataset; documents model limitations and battery-safety considerations.
Cost	Uses budget-conscious components, including ESP32-S3 and INA226.
Standards	IEC 62660-1 and IEEE 1725 are relevant references; no certification claim is made.